COURSE END PROJECT

OFFLINE MEDICINE REMINDER AND ALERT SYSTEM

IOT LABORATORY (A8603)

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

SUBMITTED BY

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VARDHAMAN COLLEGE OF ENGINEERING
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Shamshabad – 501 218, Hyderabad

2025 - 2026

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

This is to certify that the Course End Project titled **"Offline Medicine Reminder and Alert System"** is carried out by **M.Yuvan , Suhana** and **V.Pranvi Reddy** with Roll Numbers **24881A05H1, 24881A05J6 and 24881A05J9** towards **A8603 – Internet of Things Laboratory** course in partial fulfilment of the requirements for the award of degree of **Bachelor of Technology** in **Computer Science and Engineering** during the Academic year 2025-26.

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Course End Project

IOT Laboratory (A8603)

Check List

S. No.	Content	Status (✓ / ×)
1	Abstract	✓
2	Objectives	✓
3	Problem Statement	✓
4	Algorithm	✓
5	Flow Chart	✓
6	Source Code	✓
7	Input	✓
8	Output	✓
9	Conclusion	✓
10	Future Scope	✓
11	References	✓

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1.ABSTRACT

The Offline Medicine Reminder and Alert System is designed to assist patients in following their prescribed medication schedules accurately, without depending on internet connectivity or smartphones. The prototype is built around a microcontroller platform integrated with a real-time clock module, keypad, non-volatile memory, buzzer, and LCD display, allowing users or caretakers to configure multiple medicine timings, names, and dosage information directly on the device. Once the schedule is stored, the system continuously tracks the current time and, at each preset interval, generates audio-visual alerts: the buzzer sounds while the LCD shows the corresponding medicine details, dosage, and time.

The user can acknowledge the reminder using the keypad, after which the alert is silenced and the next scheduled time is awaited; if no acknowledgement is received within a defined grace period, the system repeats or extends the alert to reduce the chance of missed doses. Because all data and logic are stored locally, the device operates reliably in rural or low-connectivity environments and is suitable for elderly or medically vulnerable users who may not be comfortable with mobile applications. By offering a low-cost, standalone, and easy-to-use solution for medication management, the Offline Medicine Reminder and Alert System aims to improve adherence, prevent complications arising from skipped or mistimed doses, and support better long-term disease management for patients and caregivers alike.

2. PROBLEM STATEMENT

A significant number of patients fail to take their medicines exactly as prescribed, either forgetting doses completely or taking them at the wrong time of day. This problem is especially common among elderly people, patients with multiple chronic diseases, and individuals who must follow complex schedules involving several medicines with different timings and dosages. Poor adherence directly reduces the effectiveness of treatment, leads to uncontrolled symptoms, increases the risk of complications and hospital readmissions, and ultimately raises healthcare costs for both families and the healthcare system.

Although many reminder solutions exist, a large proportion of them are implemented as smartphone applications or cloud-based platforms that depend on continuous internet access, push notifications, and regular user interaction with a mobile device. These approaches are not ideal for users in rural or resource-constrained environments where network connectivity is unreliable, electricity supply is unstable, or smartphone penetration is low. They are also unsuitable for elderly patients who may not be comfortable using touchscreens, installing apps, or managing frequent software updates. In addition, simple tools like paper calendars or reminder cards rely heavily on user discipline and do not provide strong, attention-grabbing alerts when a dose is due.

Therefore, there is a clear need for a dedicated, low-cost, offline medicine reminder and alert system that operates independently of the internet and smartphones, but still provides robust, easy-to-understand notifications. The required system should allow caretakers or users to store multiple medicine schedules locally, specifying the time, name, and dosage of each medicine, and should automatically track the current time to generate audio-visual alerts when a dose is due. The device must be simple enough for elderly and non-technical users to operate, reliable enough to run continuously, and flexible enough to handle changes in prescription without complex reconfiguration. Addressing this problem can significantly improve medication adherence, enhance patient safety, and support better long-term management of chronic conditions in both urban and rural settings.

3. OBJECTIVES

The primary objective of the Offline Medicine Reminder and Alert System is to develop a simple, low-cost hardware device that reliably reminds patients to take medicines on time without requiring internet connectivity or smartphones. It targets elderly and chronically ill users who struggle with complex medication schedules, aiming to improve adherence and reduce missed or delayed doses through clear audio-visual alerts and an easy-to-use interface

- To store multiple medicine schedules (time, name, dosage) locally using basic keypad or button inputs.
- To generate buzzer and LCD-based alerts at each scheduled dose time and repeat them if not acknowledged.
- To provide a simple way for users or caregivers to add, edit, or delete medicine timings as prescriptions change.
- To ensure reliable, long-duration operation with minimal maintenance, suitable for home or rural healthcare settings.

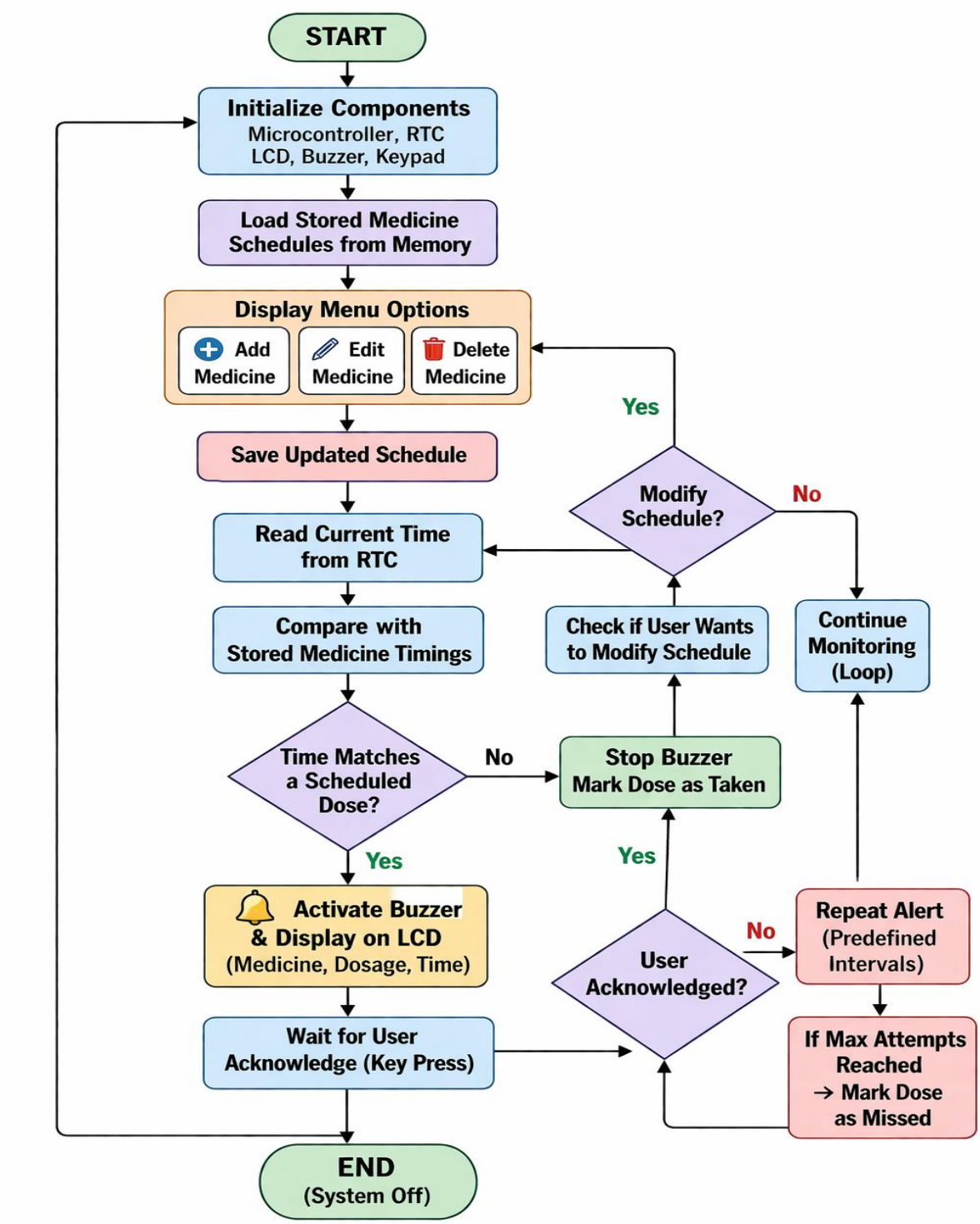
4. Algorithm

Algorithm for the Offline Medicine Reminder and Alert System:

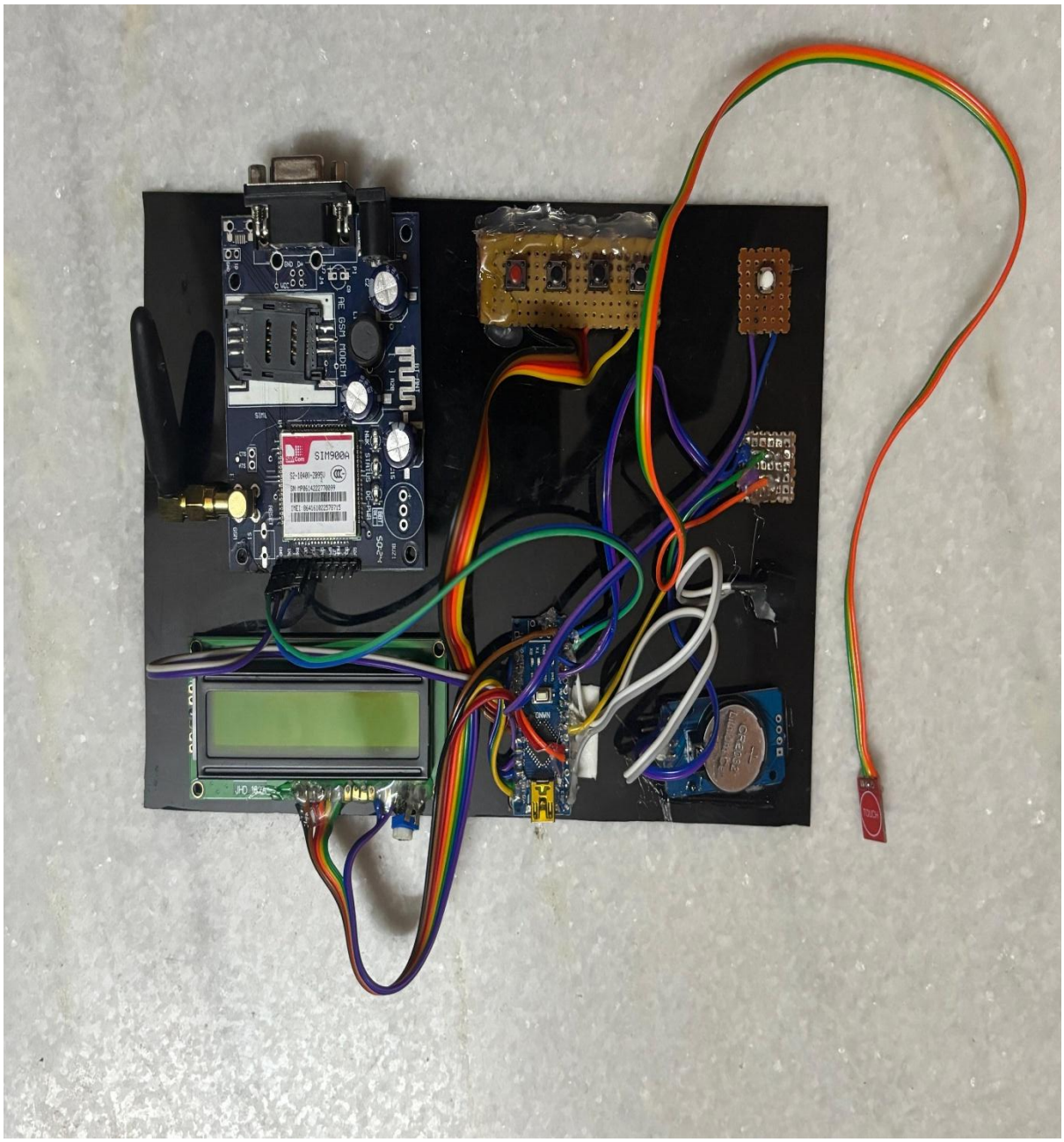
1. Start.
2. Initialize the microcontroller, real-time clock (RTC), LCD, buzzer, and input keys.
3. Load any previously saved medicine schedules from non-volatile memory.
4. Provide a menu for the user or caregiver to add, edit, or delete medicine entries (name, time, dosage) and save them.
5. Continuously read the current time from the RTC.
6. Compare the current time with all stored medicine timings.
7. If a match is found for a dose that is not yet acknowledged:
    - Turn on the buzzer and display the corresponding medicine details on the LCD.
    - Wait for an acknowledgement key press within a predefined reminder interval.
8. If the user acknowledges within the interval, stop the buzzer, mark the dose as taken, and return to time monitoring.
9. If there is no acknowledgement, repeat the alert a set number of times and then record the dose as missed.
10. Regularly check whether the user wants to modify the schedule; if yes, return to the menu step.
11. End (on device power-off).

5. PROPOSED SYSTEM

5.1 FLOW CHART



5.2WORKING MODEL



6.SOURCE CODE

```
#include <Wire.h>

#include <RTClib.h>

#include <LiquidCrystal.h>

#include <EEPROM.h>


// RTC and LCD

RTC_DS3231 rtc;

LiquidCrystal lcd(7, 8, 9, 10, 11, 12);


// Buzzer

int buzzer = 6;


// Medicine structure

struct Medicine {

  int hour;

  int minute;

  char name[10];

  int taken; // 0 = not taken, 1 = taken

};


// Max medicines

#define MAX_MED 5

Medicine meds[MAX_MED];
```

```
int medCount = 0;

// Function prototypes
void loadData();
void saveData();
void checkMedicine();
void alertUser(Medicine &m);

void setup() {
  Serial.begin(9600);
  Wire.begin();
  rtc.begin();

  lcd.begin(16, 2);
  pinMode(buzzer, OUTPUT);

  lcd.print("Medicine System");
  delay(2000);
  lcd.clear();

  loadData();
}

void loop() {
```

```
  DateTime now = rtc.now();

  lcd.setCursor(0, 0);
  lcd.print("Time: ");
  lcd.print(now.hour());
  lcd.print(":");
  lcd.print(now.minute());

  checkMedicine();

  delay(1000);
}

// Load data from EEPROM
void loadData() {
  EEPROM.get(0, medCount);
  if (medCount > MAX_MED) medCount = 0;

  for (int i = 0; i < medCount; i++) {
    EEPROM.get(10 + i * sizeof(Medicine), meds[i]);
  }
}

// Save data to EEPROM
```

```
void saveData() {
  EEPROM.put(0, medCount);
  for (int i = 0; i < medCount; i++) {
    EEPROM.put(10 + i * sizeof(Medicine), meds[i]);
  }
}

// Check time with medicines
void checkMedicine() {
  DateTime now = rtc.now();

  for (int i = 0; i < medCount; i++) {
    if (meds[i].hour == now.hour() &&
        meds[i].minute == now.minute() &&
        meds[i].taken == 0) {

      alertUser(meds[i]);
    }
  }
}

// Alert logic
void alertUser(Medicine &m) {
  int attempts = 0;
```

```
while (attempts < 3) {
  lcd.clear();
  lcd.print("Take: ");
  lcd.print(m.name);

  lcd.setCursor(0, 1);
  lcd.print("Press key!");

  digitalWrite(buzzer, HIGH);
  delay(1000);
  digitalWrite(buzzer, LOW);

  // Simulated key press via Serial
  if (Serial.available()) {
    char input = Serial.read();
    if (input == '1') {
      m.taken = 1;
      lcd.clear();
      lcd.print("Dose Taken");
      delay(2000);
      return;
    }
  }
```

```
    attempts++;
    delay(3000);
  }

  // If not acknowledged
  lcd.clear();
  lcd.print("Dose Missed");
  delay(2000);
}
```

7.INPUT

1. Current date and time from the real-time clock (RTC).
2. Medicine details entered by the user/caregiver: medicine name, dosage, and scheduled intake time(s).
3. User key presses for menu navigation (add/edit/delete), acknowledgement of reminders, and repetition or cancellation of alerts.

8.CONCLUSION

The Offline Medicine Reminder and Alert System successfully demonstrates a practical solution for improving medication adherence using a simple stand-alone hardware design. By combining a real-time clock, microcontroller, buzzer, and LCD interface, the prototype can store multiple medicine schedules locally and generate clear audio-visual alerts at the exact time a dose is due, without relying on smartphones or internet connectivity. This makes the system particularly suitable for elderly patients and users in rural or low-connectivity areas, where app-based or cloud-based reminder solutions are difficult to adopt.

Through its configurable scheduling, acknowledgement, and repeated alert features, the device directly addresses the common problem of missed or delayed doses and supports more consistent, on-time medication intake. As a result, it has the potential to reduce preventable complications arising from poor adherence and to enhance the quality of life for chronically ill patients and their caregivers. The project also shows that relatively low-cost embedded hardware can be effectively used to tackle real-world healthcare challenges, providing a strong foundation for future enhancements such as dose logging, battery-backup, and optional integration with mobile or GSM modules for caretaker notifications.

9.FUTURE SCOPE

The Offline Medicine Reminder and Alert System has strong potential for further enhancement into a more comprehensive medication-management solution. By adding communication, sensing, and data-logging capabilities, the device can move beyond basic reminders and provide richer support for patients, caregivers, and healthcare professionals.

Future scope includes:

- Integrating GSM/Bluetooth/Wi-Fi modules to send SMS or app notifications to caregivers when a dose is missed or when the device battery is low.
- Converting the hardware into a compartment-based pill box with lid-open detection so the correct medicine is taken at the correct time.
- Storing detailed dose history in external memory or syncing it to a companion app for generating adherence reports for doctors.
- Adding simple health-monitoring sensors (e.g., heart-rate, blood-pressure) to record basic vitals along with medication events.
- Providing advanced user features such as multilingual voice prompts, adjustable reminder patterns for highly forgetful users, and optional integration with electronic health records.

10.REFERENCES

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